

ARNAV GOEL

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EDUCATION

National University of Singapore

Aug 2023 - May 2027 [Expected]

Bachelor of Computing in Computer Science (Honours)
Specialising in Artificial Intelligence and Cybersecurity

GPA: 4.14/5.00

Final Year Project: Building a **watchOS/iOS palliative-care monitoring system** using **explainable machine learning** on wearable vitals and movement signals to identify early deterioration with **Apple & National University Hospital (NUH)**

Teaching Assistant: Introduction to Software Engineering and Spatial Computing: Design and Development

Global Experience: NUS Global Internship Programme in Bangkok, Thailand

TECHNICAL & LANGUAGE SKILLS

Programming Languages: Python, Java, Swift, SQL, TypeScript, JavaScript, Kotlin, R

AI/ML: PyTorch, RAG, Agentic Workflows, LangGraph, MCP, Machine Learning, Natural Language Processing, OpenCV

Web Development & Database: React.js, Next.js, Node.js, REST APIs, MySQL, PostgreSQL, Firebase, Material UI

Tools & Infrastructure: AWS, Docker, Git, CI/CD, GitHub Actions, Cloudflare, Dataiku, Tableau, Figma, Xcode

Native Proficiency: English and Hindi

WORK EXPERIENCE

Johnson & Johnson, Singapore

Jan 2026 – Jul 2026

Data & Digital Technology Analyst Intern [2 Appreciation Awards]

- Built a **Graph RAG system** that cut sales information search and validation **from 30 minutes to <1 minute**, using Qdrant **vector search** and end-to-end **observability** to deliver evidence-backed answers with traceable citations.
- Engineered an **AI workflow platform for Business Analysts** designed to cut product planning and documentation from **4 weeks to <3 days**, generating user flows, stories and editable mockups with automatic document updates.
- Created **working AR-glasses and mobile prototypes** as scalable solutions designed to integrate with existing business workflows, and developed **Dataiku ETL** automations for inventory management.

Interactive 3D, Singapore

May 2025 – Jan 2026

Software Engineer Intern

- Led a 3-person team with NUH to build an **Apple Vision Pro/iPad stroke-rehabilitation system** using Swift, transforming traditional visual assessments into **quantified, real-time movement metrics** to track patient recovery.
- Developed a **patent-pending 3D hand-dexterity assessment**, refining it through testing with **7 patients** and clinician feedback to improve usability and align with real-world rehabilitation workflows.
- Built a **Python analytics pipeline** using Pandas and Matplotlib to process **15,000+ patient data points**, transforming raw assessment data into visual trends for longitudinal recovery analysis.

Institute for Applied Sciences and Educational Technology (NUS), Singapore

Nov 2023 – Jun 2024

Software Engineer Intern

- Built a **full-stack coursework management platform** that **reduced administrative workload by 30%**, using React.js, JavaScript and MySQL to build real-time progress tracking and end-to-end CRUD workflows.
- Designed a **normalized MySQL relational database** and optimized indexed multi-table queries, **improving report-generation speed by ~4x** across 10+ interconnected tables while maintaining data integrity.

PROJECTS

Multi-Agent F&B Procurement: Created a **stateful multi-agent procurement system** that cut manual coordination effort per order from **~90 minutes to ~15 minutes**, automating inventory requests, supplier negotiation and invoice verification across Gmail, PostgreSQL and Telegram using LangGraph/LangChain.

Agentic Shopping & Payments: Engineered a **multi-agent shopping and payments platform** that enabled a single mobile agent to **search, compare and purchase products across Shopee and Amazon**, using AWS AgentCore Browser, merchant-specific MCP tools, and programmable payments through disposable virtual cards.

Privacy-First Mobile App: Designed a **novel Android live-streaming app** using Java, **on-device computer vision and OCR** to protect sensitive content in real time by masking faces, QR codes, private text, and custom regions before RTMP transmission to social media platforms.

3D Local Discovery Platform: Launched a public web platform serving **100+ monthly users**, engineering web-scraping pipelines for events, restaurants and live promotions using OneMap and **TinyFish**, and integrating the **OpenAI Realtime API** for voice-based natural-language discovery.

EXTRA-CURRICULAR AND ADDITIONAL INFORMATION

Leadership Experience

- Appointed as **Prince George's Park Residence Cluster Leader**, responsible for hosting events with 15 floor residents and overseeing administrative duties such as resident safety and fire drills.
- **Co-organised the world's first university hackathon** dedicated to Apple Vision Pro development, managing website development and mentoring student participants throughout the event.
- Served as **Press and Creative Writing Executive for the NUS Students' Union**, crafting LinkedIn captions for events and initiatives, while developing article topics and weekly fun facts.

Volunteering Experience

- Volunteering with **NUS TeachSG**, working with Unity Secondary School to mentor secondary school students and support their learning and development.
- Acted as **Vice-Project Director for the Elderly Service Project** (Computing Club), delegating tasks to various subdepartments, organising NUH volunteering sessions, while managing proposal writing and user guides for elderly.
- Volunteered as a **Python/Scratch Instructor for CITC** and a **Digital Literacy Instructor for SHG**, teaching programming basics to students and guiding seniors in app usage, honing communication and instructional skills.

Achievements

- Achieved a **top 10 placement in a national creative writing competition**, leading to the publication of my debut book, *Trouble At Thames*, now stocked in every **National Library Board branch** across Singapore.

Hobbies

- Landscape photographer, certified guitarist and passionate Tour de France fan.